

A shock-capturing numerical scheme for a non-conservative self-organized hydrodynamics model



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Physical model

This work is devoted to the numerical approximation of the weak solutions of the **Self-Organized Hydrodynamics (SOH)** model, defined in 1D as follows :

$$\partial_t \rho + \partial_x(\rho u) = 0, \quad (1a)$$

$$\rho(\partial_t u + cu \partial_x u) + \lambda(1 - u^2) \partial_x \rho = 0, \quad (1b)$$

$$\rho(\partial_t v + cu \partial_x v) - \lambda uv \partial_x \rho = 0, \quad (1c)$$

$$\sqrt{u^2 + v^2} = 1. \quad (1d)$$

This system models a **collective self-organized motion** where $\rho > 0$ is the density of individuals, $\Omega = (u, v)^T$ the speed of the flow, and $0 < c < 1$ and $\lambda > 0$ two given constants. These behaviours can be observed in nature as insect swarms, fish schools or bird flocks for example.

Objectives

The SOH model (1) is **hyperbolic** and **non-conservative**.

- ★ Shock wave solutions are **not well-defined**. → They are defined as limits of traveling waves solutions of the viscous SOH model [1].
- ★ Non-conservative schemes are claimed to **converge to wrong solutions**.

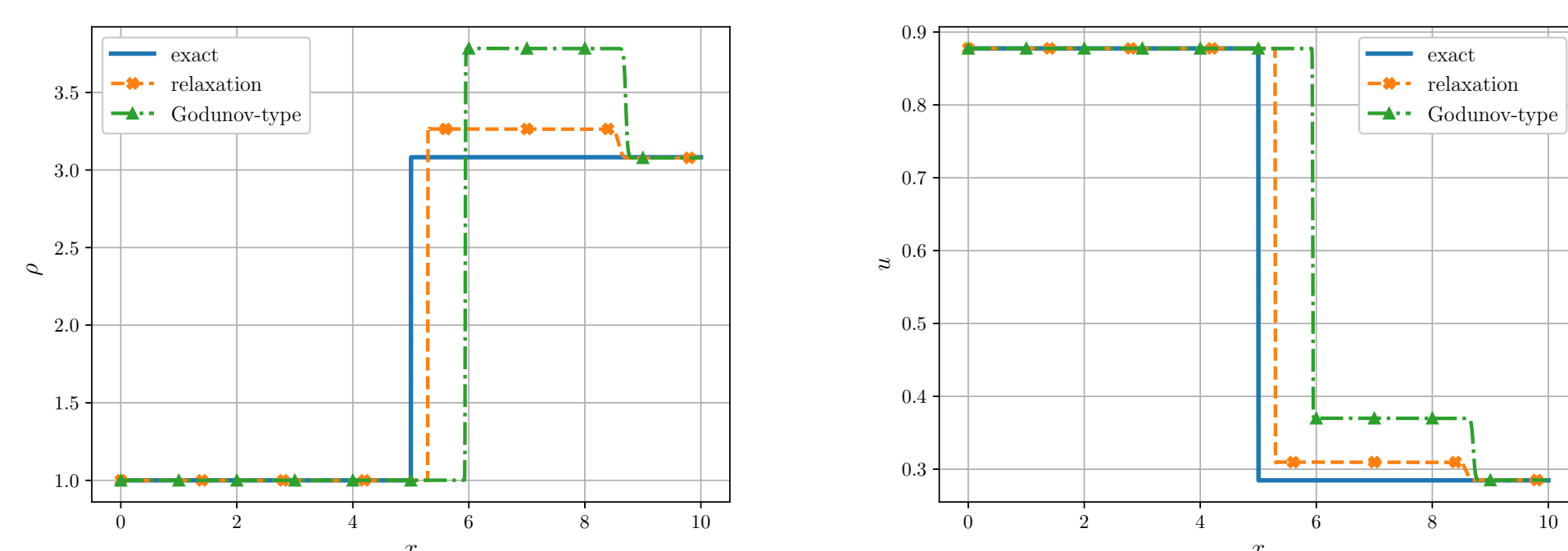


Figure – Stationary shock at $T = 5$ with 5000 cells. In blue : exact solution. In orange : relaxation method [2]. In green : Godunov-type scheme without viscous term.

→ A viscous correction is added to the Godunov-type scheme.

Reformulation of the model

For regular and shock wave solutions, the SOH model (1) satisfies the geometric constraint as long as the initial velocity is well-prepared, i.e. $\Omega_0 \in \mathbb{S}^1$. Considering the change of variable $\Omega = (\cos \theta, \sin \theta)^T$, the SOH model (1) recasts as

$$\partial_t \rho + \partial_x(\rho \cos \theta) = 0, \quad (2a)$$

$$\partial_t \theta + \partial_x(c \sin \theta) - \lambda \frac{\sin \theta}{\rho} \partial_x \rho = 0, \quad (2b)$$

Traveling waves

We consider a non-conservative hyperbolic PDE system defined as follows :

$$\partial_t w + A(w) \partial_x w = 0. \quad (3)$$

Its **associated viscous model** with a rescaling reads :

$$\partial_t w^\varepsilon + A(w^\varepsilon) \partial_x w^\varepsilon = \varepsilon D_0(w^\varepsilon) \partial_x (D_1(w^\varepsilon) \partial_x w^\varepsilon), \quad (4)$$

where $\varepsilon > 0$ and $\varepsilon \rightarrow 0$.

Definition Traveling waves are solutions of (4) defined as follows :

$$w^\varepsilon(x, t) = \hat{w}\left(\frac{x - \sigma t}{\varepsilon}\right), \quad (5)$$

where σ is the wave speed and

$$\lim_{\xi \rightarrow -\infty} \hat{w}(\xi) = w_L \text{ and } \lim_{\xi \rightarrow +\infty} \hat{w}(\xi) = w_R. \quad (6)$$

→ A shock wave w solution of (3) compatible with the viscosity (D_0, D_1) is the **limit when $\varepsilon \rightarrow 0$ of a traveling wave $\hat{w}$** solution of (4).



Figure – Viscous profile of a shock.

References

- [1] P. DEGOND et al. “Macroscopic models of collective motion and self-organization”. In : *Séminaire Laurent Schwartz — EDP et applications* (2014).
- [2] S. MOTSCH et L. NAVORET. “Numerical Simulations of a Nonconservative Hyperbolic System with Geometric Constraints Describing Swarming Behavior”. In : *Multiscale Modeling & Simulation* (2011).

Shock wave solutions of the SOH model

The **viscous SOH system** [1] recasts under the form of (4) where

$$w = \begin{pmatrix} \rho \\ \theta \end{pmatrix}, \quad A(w) = \begin{pmatrix} \cos \theta & -\rho \sin \theta \\ -\lambda \frac{\sin \theta}{\rho} & c \cos \theta \end{pmatrix} \quad D_0(w) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\rho^2} \end{pmatrix}, \text{ and } D_1(w) = \begin{pmatrix} 0 & 0 \\ 0 & \rho^2 \end{pmatrix}.$$

Generalized Rankine-Hugoniot conditions For $\sigma \neq \cos \theta$, let us introduce

$$G'_\sigma(\theta) = \frac{-\sigma + c \cos \theta - \lambda \frac{\sin^2 \theta}{\cos \theta - \sigma}}{(\cos \theta - \sigma)^2}. \quad (7)$$

Given two initial states $w_L, w_R \in \mathcal{D} = \mathbb{R}_+^* \times (-\pi, \pi]$ and a wave speed σ , suppose that :

(a) w_L, w_R and σ verify

$$-\sigma(\rho_R - \rho_L) + (\rho_R \cos \theta_R - \rho_L \cos \theta_L) = 0, \quad (8a)$$

$$G_\sigma(\theta_R) - G_\sigma(\theta_L) = 0. \quad (8b)$$

(b) if $\theta_L < \theta_R$, for all $\theta \in (\theta_L, \theta_R)$, $G_\sigma(\theta) - G_\sigma(\theta_L) > 0$,
if $\theta_L > \theta_R$, for all $\theta \in (\theta_R, \theta_L)$, $G_\sigma(\theta) - G_\sigma(\theta_L) < 0$.

(c) for all θ between θ_L and θ_R , $\cos \theta \neq \sigma$.

Then w_L and w_R define a **shock wave** of speed σ compatible with the SOH viscous model.

Godunov-type scheme

We use the following **approximate Riemann solver** :

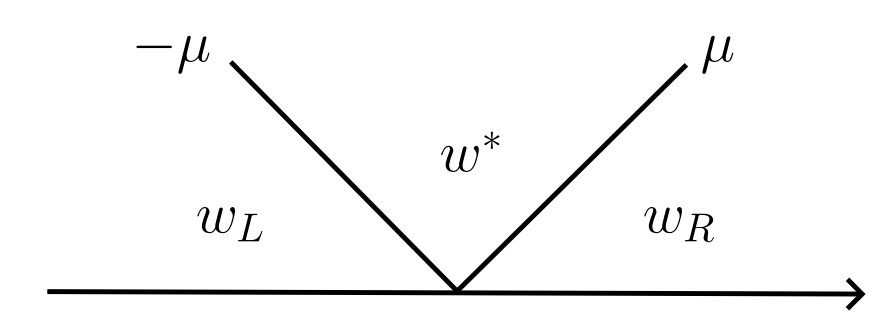


Figure – Illustration of the approximate Riemann solver.

where w^* is the intermediate state determined such that :

$$\frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \tilde{w}_R \left(\frac{x}{\Delta t}; w_L, w_R \right) dx = \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} w_R^{\text{ex}} \left(\frac{x}{\Delta t}; w_L, w_R \right) dx,$$

w_R^{ex} is the exact Riemann solver and $\mu = \max(|\mu_\pm(\theta_L)|, |\mu_\pm(\theta_R)|)$.

→ The approximate Riemann solver is used at each interface of the mesh.

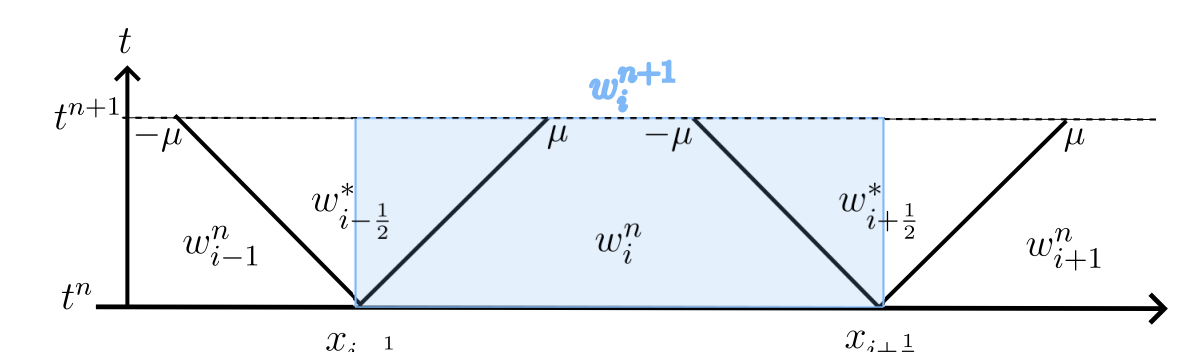


Figure – Juxtaposition of Riemann solvers.

Godunov-type scheme

$$w_i^{n+1} = w_i^n - \frac{\Delta t}{\Delta x} \left(f(w_i^n, w_{i+1}^n) - f(w_{i-1}^n, w_i^n) + \frac{1}{2} (a_{i-1/2} + a_{i+1/2}) \right),$$

where

$$f(w_L, w_R) = \begin{pmatrix} \frac{1}{2}(\rho_L \cos \theta_L + \rho_R \cos \theta_R) - \frac{\mu}{2}(\rho_R - \rho_L) \\ \frac{c}{2}(\sin \theta_L + \sin \theta_R) - \frac{\mu}{2}(\theta_R - \theta_L) \end{pmatrix}$$

$$\text{and } a_{LR} = \begin{pmatrix} 0 \\ -\lambda \frac{\sin \theta_{LR}}{\rho_{LR}}(\rho_R - \rho_L) \end{pmatrix}.$$

Viscous correction

Idea : Shock waves are defined thanks to the **physical viscosity** of the SOH model → this viscosity is discretized and added to the Godunov-type scheme. The scheme on θ then becomes

$$\theta_i^{n+1} = \theta_i^n - \frac{\Delta t}{\Delta x} \left(f^\theta(w_i^n, w_{i+1}^n) - f^\theta(w_{i-1}^n, w_i^n) + \frac{1}{2} (a_{i-1/2}^\theta + a_{i+1/2}^\theta) \right) - \alpha_{HLL} V_{HLL,i}^n + \alpha_{phys} V_{phys,i}^n,$$

where $V_{HLL,i}^n$ is the **natural viscosity of the Godunov-type scheme** and $V_{phys,i}^n$ is a discretization of the **physical viscosity** of the viscous SOH model.

Numerical results

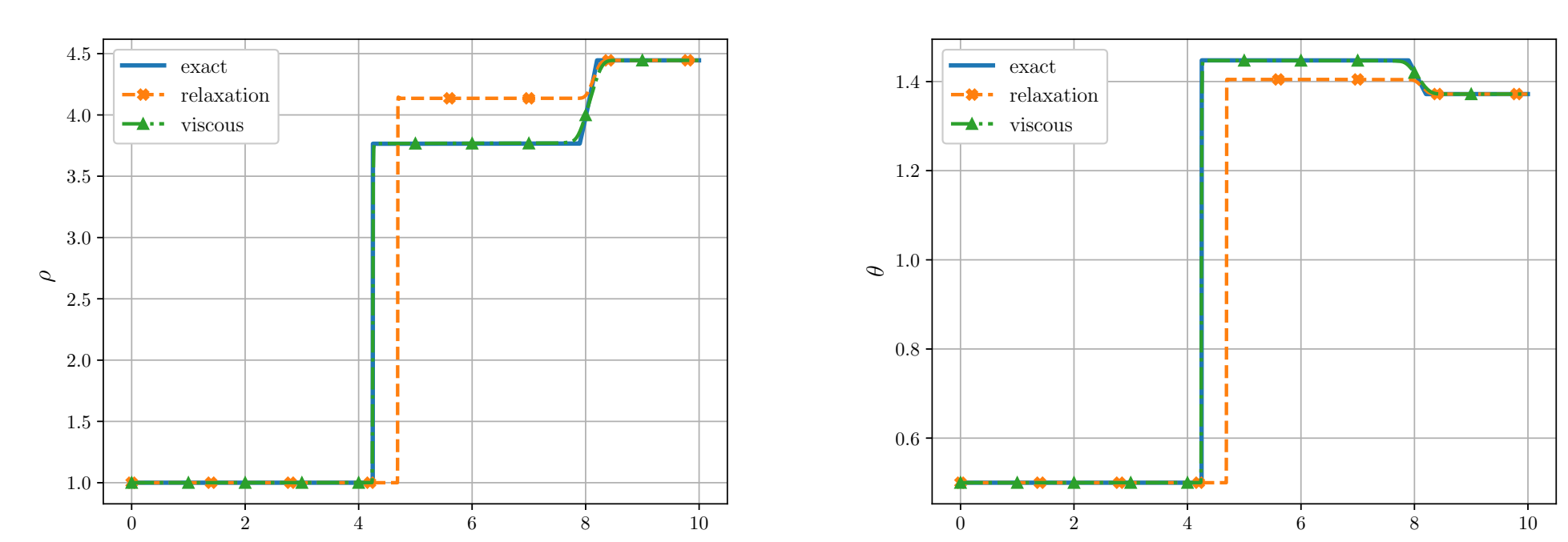


Figure – 1-shock and 2-rarefaction at time $T = 5$ with 5000 cells and $\alpha_{HLL} = 1.871$. In blue : exact solution. In orange : relaxation method [2]. In green : viscous scheme.